Manufacturing Operational Insights Report

(Finance Support Triage Agent)

Course Name: Agentic AI

Institution Name: Medicaps University – Datagami Skill Based Course

Student Name(s) & Enrolment Number(s):

Sr.	Student Name	Enrollment Number
1	JIDNYAS BHUSHAN BHONGE	EN22CS301469
2	KASHISH SHARMA	EN22CS301492
3	KRATI PATIDAR	EN22CS301514
4	KHUSHI MEHRA	EN22CS301504
5	KAMAL YADAV	EN22CS301476

Group Name: 03D5

Project Number: Group 3

Industry Mentor Name:_Mr. Sooraj Nair

University Mentor Name: Prof. Maya Yadav Baniya

Academic Year: 2025-26

Table of Contents

1. **Introduction**
   1.1 Problem Statement
   1.2 Project Objectives
2. **Scope of the Project**
3. **Proposed Solution**
4. **Key Features**
   4.1 Automated Email Triage
   4.2 Intent, Category, and Priority Classification
   4.3 Financial Entity Extraction
   4.4 Draft Response Generation
   4.5 Ticket Storage and Management
   4.6 Dashboard Visualization
5. **Overall Architecture / Workflow**
   5.1 Email Reception via Gmail (IMAP)
   5.2 Backend Processing with FastAPI
   5.3 Structured Prompt Construction using LangChain
   5.4 AI Processing using Llama 3.3
   5.5 Schema Validation using Pydantic
   5.6 Ticket Storage in PostgreSQL
   5.7 Visualization via Streamlit Dashboard
   5.8 Reply Workflow via SMTP
6. **Tools & Technologies Used**
7. **Results & Output**
   7.1 Screenshots / Outputs
   7.2 Reports / Dashboards
   7.3 Key Outcomes
8. **Conclusion**
9. **Future Scope & Enhancements**

Problem Statement & Objectives

Problem Statement:

Finance teams receive a large volume of customer support emails related to fraud reports, payment failures, refunds, and account issues. Manually reading and categorizing these emails is time-consuming, inconsistent, and prone to delays, especially in identifying high-priority or fraud-related cases.
To address this challenge, this project proposes an **Agentic AI-based Finance Support Triage System** that automatically analyzes incoming emails, classifies intent, category, and urgency, extracts key financial entities, generates draft responses, and organizes tickets through an interactive dashboard.

Project Objectives:

- Automate triaging of incoming finance support emails.
- Classify emails based on intent, category, and priority level.
- Extract structured financial entities such as transaction amount, merchant name, and issue type.
- Generate structured support tickets for tracking and analytics.
- Provide AI-generated draft replies to reduce response time.
- Enable monitoring and decision-making via an analytics dashboard.
- Highlight fraud and high-risk cases through alerts and priority queues.

Scope of the Project

The scope of this project involves the design and development of an AI-powered finance support automation system that streamlines the triaging of customer support emails. The system focuses on transforming unstructured email data into structured, actionable insights using Agentic AI and modern backend technologies.

The project includes the implementation of an intelligent backend service capable of fetching finance-related emails, processing them using a Large Language Model (LLM), and converting them into structured support tickets. It covers the complete pipeline from email ingestion and AI-based classification to data validation and storage in a relational database. The system also includes an interactive dashboard that enables visualization of categorized tickets, priority queues, alerts, and operational analytics.

Proposed Solution

The proposed solution is an end-to-end AI-driven Finance Support Triage System designed to automate the classification and management of finance-related support emails using Agentic AI principles. The system combines modern backend frameworks, Large Language Models (LLMs), and data visualization tools to create a scalable and intelligent triage pipeline.

At the core of the solution is a robust backend built using **FastAPI**, which acts as the central orchestration layer. It handles email ingestion, API routing, request validation, and communication with the AI processing pipeline. Incoming emails are fetched through Gmail IMAP integration and forwarded to the AI engine for analysis.

The AI processing layer leverages **Llama 3.3 (via Groq API)** integrated through **LangChain**, which enables structured prompt engineering and consistent output formatting. The LLM analyzes the content of each email to identify intent, categorize the issue type, assign a priority level, and extract key financial entities such as transaction amounts, merchants, and issue context. Structured outputs are validated using schema enforcement (e.g., Pydantic) to ensure reliability and consistency.

Processed data is then stored in a **PostgreSQL database**, enabling persistent storage of support tickets along with metadata such as category, urgency, extracted entities, and AI-generated draft replies. This structured storage layer allows for easy querying, analytics, and future integration with enterprise systems.

On the frontend, a lightweight yet powerful **Streamlit dashboard** is used to visualize and manage triaged tickets. The dashboard provides multiple operational views such as inbox categorization, priority queues, alerts for fraud detection, and analytics summaries. It also supports workflows for reviewing AI-generated draft responses, enabling human-in-the-loop decision-making before sending replies.

Overall, the proposed solution delivers a modular and scalable architecture that demonstrates how Agentic AI can be effectively integrated into real-world support operations. It reduces manual workload, improves response time, and provides actionable insights through structured automation and interactive visualization.

Key Features

1. Automated Email Triage

The system automatically reads incoming finance-related emails and processes them without manual intervention. Instead of support agents manually reviewing each email, the AI engine analyzes the content and routes it into appropriate categories. This significantly reduces processing time and minimizes human error, especially when handling large volumes of emails daily.

2. Intent, Category, and Priority Classification

Using a Large Language Model (LLM), the system identifies the **intent** of the email (e.g., refund request, fraud complaint, payment failure), classifies it into predefined **categories**, and assigns a **priority level** (High, Medium, Low).

- High priority: Fraud, unauthorized transactions
- Medium priority: Payment delays, refund issues
- Low priority: General queries

This intelligent classification ensures urgent cases are handled first and resources are allocated efficiently.

3. Financial Entity Extraction

The system extracts important financial information from unstructured email text and converts it into structured data. These entities may include:

- Transaction amount
- Merchant name
- Date of transaction

- Payment method
- Account reference details

This structured extraction enables better tracking, reporting, and analytics while eliminating manual data entry.

4. Draft Response Generation

The AI model generates context-aware draft replies based on the classified issue. These drafts help support agents respond faster while maintaining consistency in communication. A human-in-the-loop mechanism allows agents to review and approve responses before sending them, ensuring quality control.

5. Ticket Storage and Management

Once processed, each email is converted into a structured support ticket and stored in a PostgreSQL database. The stored data includes:

- Email content
- Intent and category
- Priority level
- Extracted entities
- AI-generated draft reply
- Status (Open / In Progress / Resolved)

This structured storage system enables easy ticket tracking, reporting, and future system integrations.

6. Dashboard Visualization

A Streamlit-based dashboard provides real-time visibility into the triaged tickets. It includes:

- Categorized ticket views

- Priority queues
- Fraud alerts
- Summary analytics (e.g., ticket distribution by category)

The dashboard helps decision-makers monitor operational performance and quickly identify high-risk cases.

Overall Architecture / Workflow

The system follows a modular and layered architecture that integrates email ingestion, AI processing, structured data storage, and visualization. The workflow ensures smooth automation from receiving an email to generating actionable insights and draft responses.

1. Email Reception via Gmail (IMAP)

The workflow begins when a customer sends a finance-related support email. The system connects to the Gmail inbox using **IMAP integration**, which enables automated email fetching. This eliminates manual monitoring and ensures continuous ingestion of incoming support requests.

2. Backend Processing with FastAPI

Once an email is received, it is forwarded to the backend service built using **FastAPI**. This backend acts as the central controller of the system and handles:

- Request routing
- Input preprocessing
- API orchestration
- Integration with AI components

FastAPI ensures high performance and scalability while maintaining clean API structure.

3. Structured Prompt Construction using LangChain

The backend passes the email content to **LangChain**, which is responsible for building structured prompts for the Large Language Model (LLM). LangChain helps enforce consistency in the output by guiding the model to produce predefined structured formats rather than free-form text.

4. AI Processing using Llama 3.3

The processed prompt is sent to **Llama 3.3 (via Groq API)**, which performs the core intelligence of the system. The model analyzes the email content and generates structured insights such as:

- Intent detection
- Category classification
- Priority tagging
- Financial entity extraction
- Suggested draft response

The output is generated in structured JSON format for downstream processing.

5. Schema Validation using Pydantic

To ensure reliability and correctness, the generated JSON output is validated using **Pydantic schemas**. This step ensures that all required fields (e.g., priority, category, entities) are present and correctly formatted, reducing errors and improving data consistency.

6. Ticket Storage in PostgreSQL

After validation, the structured data is stored in a **PostgreSQL database**. Each processed email becomes a support ticket containing metadata such as

classification, urgency, extracted entities, and AI-generated replies. This storage layer enables persistent tracking and analytics.

7. Visualization via Streamlit Dashboard

The stored ticket data is visualized using a **Streamlit dashboard**, which provides an interactive interface for support teams. The dashboard includes:

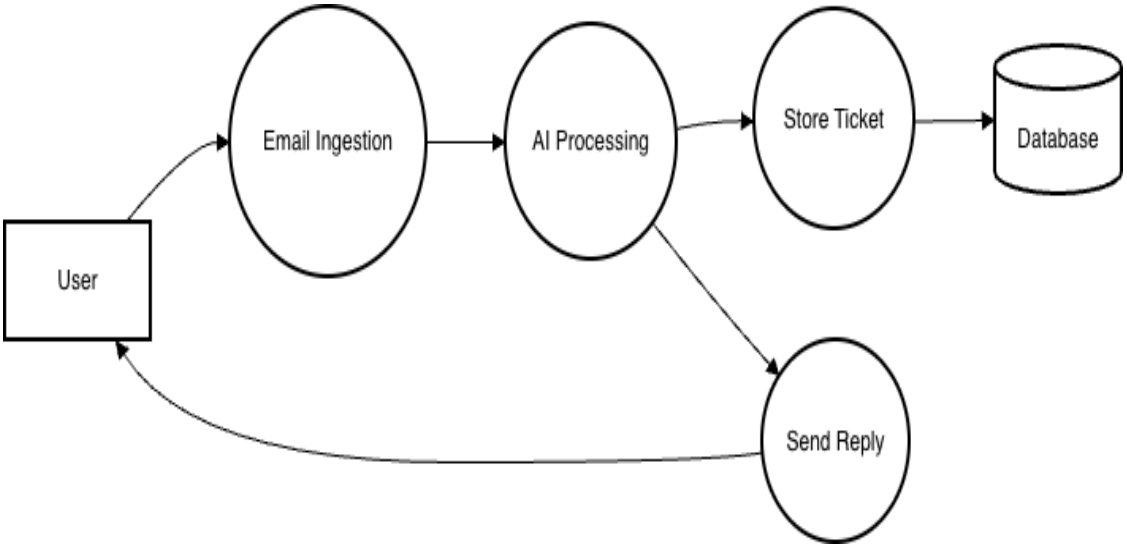
- Categorized ticket views
- Priority queues
- Fraud alerts
- Operational analytics

This enables quick monitoring and data-driven decision-making.

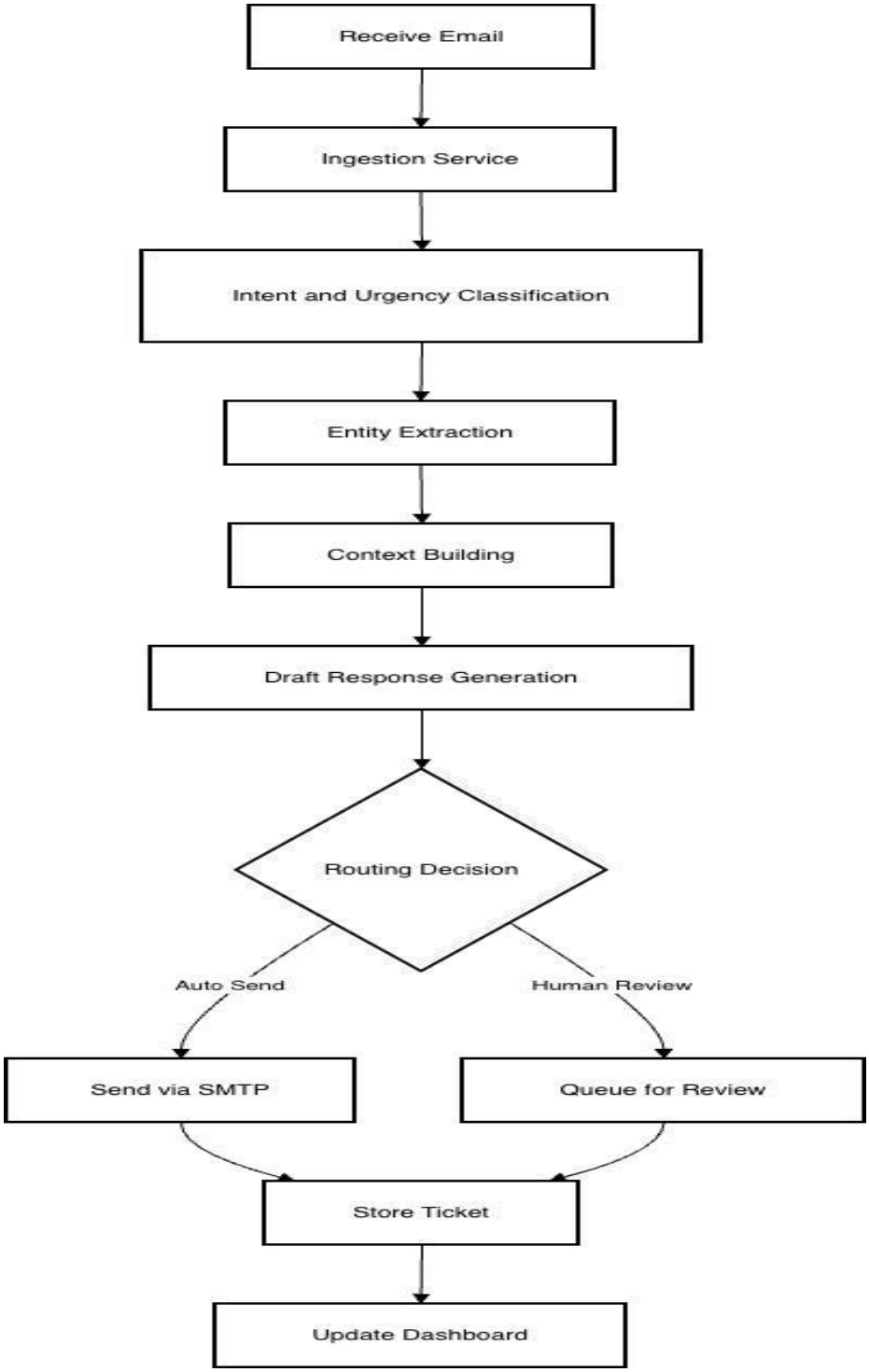
8. Reply Workflow via SMTP

For response handling, the system integrates **SMTP-based email sending**. AI-generated draft replies can be reviewed by agents and then sent to customers through an approval workflow. This ensures faster communication while maintaining human oversight.

Information flow Diagram:



Process Flow Diagram:



Tools & Technologies Used

- Backend: Python, FastAPI
- AI Model: Llama 3.3 70B (Groq API)
- AI Framework: LangChain
- Database: PostgreSQL + SQLAlchemy
- Frontend: Streamlit Dashboard
- Email: Gmail IMAP + SMTP
- OCR: EasyOCR
- Validation: Pydantic

Results & Output

Screenshots / Outputs:

- Dashboard showing ticket entries.
- Structured JSON output of processed emails.

Figure 1:

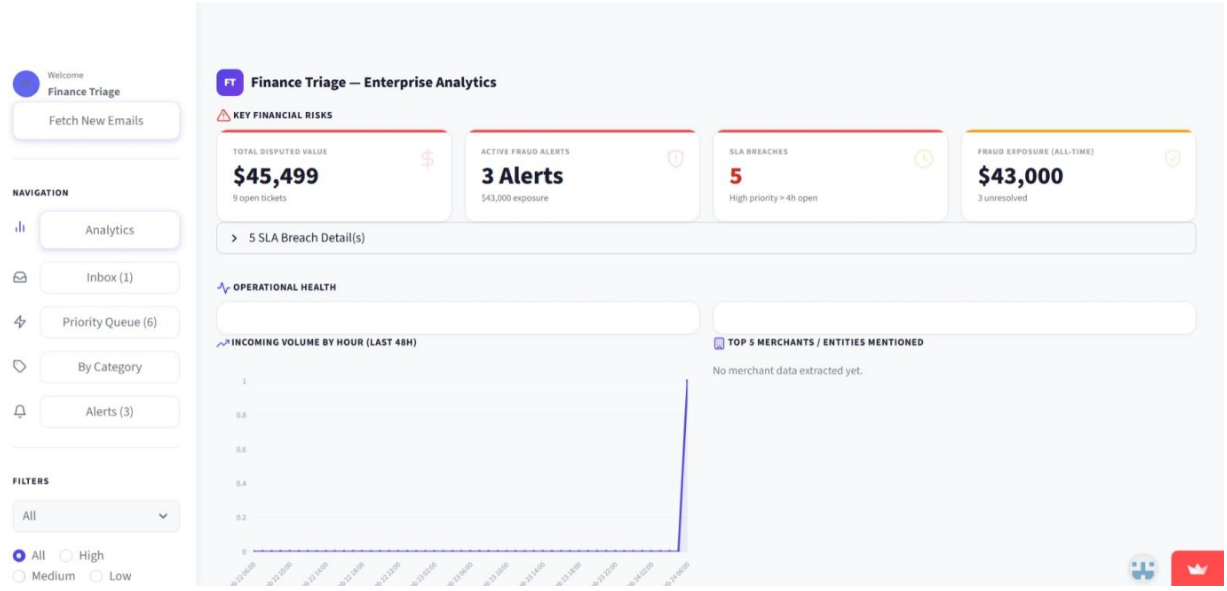


Figure 2:

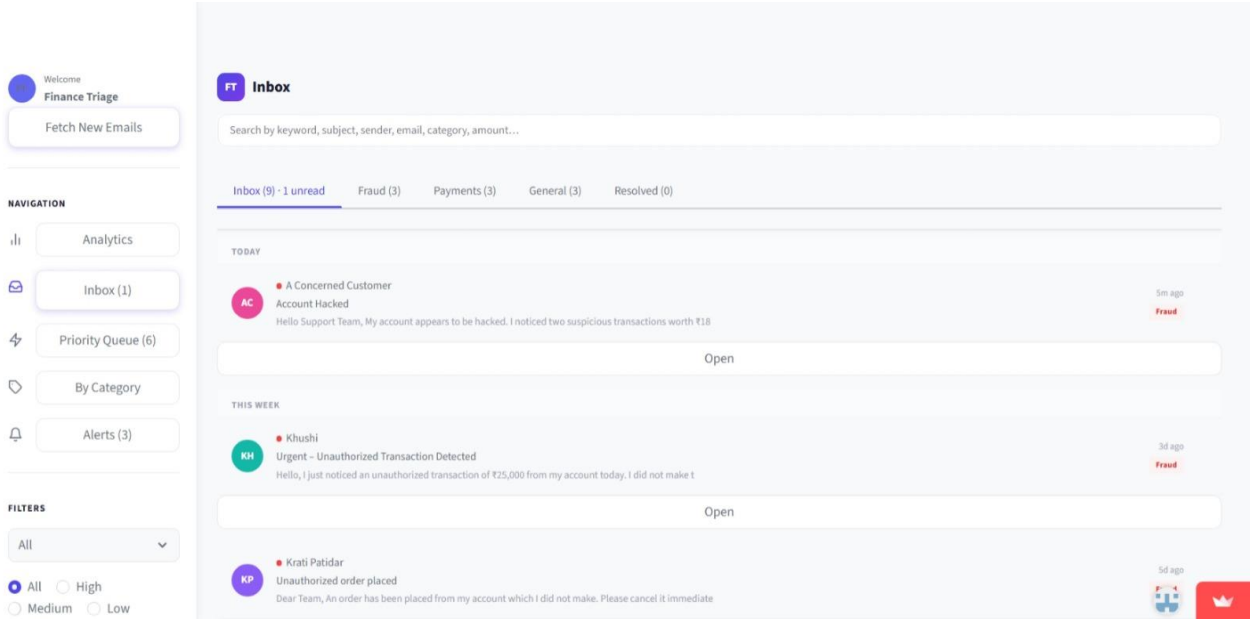


Figure 3:

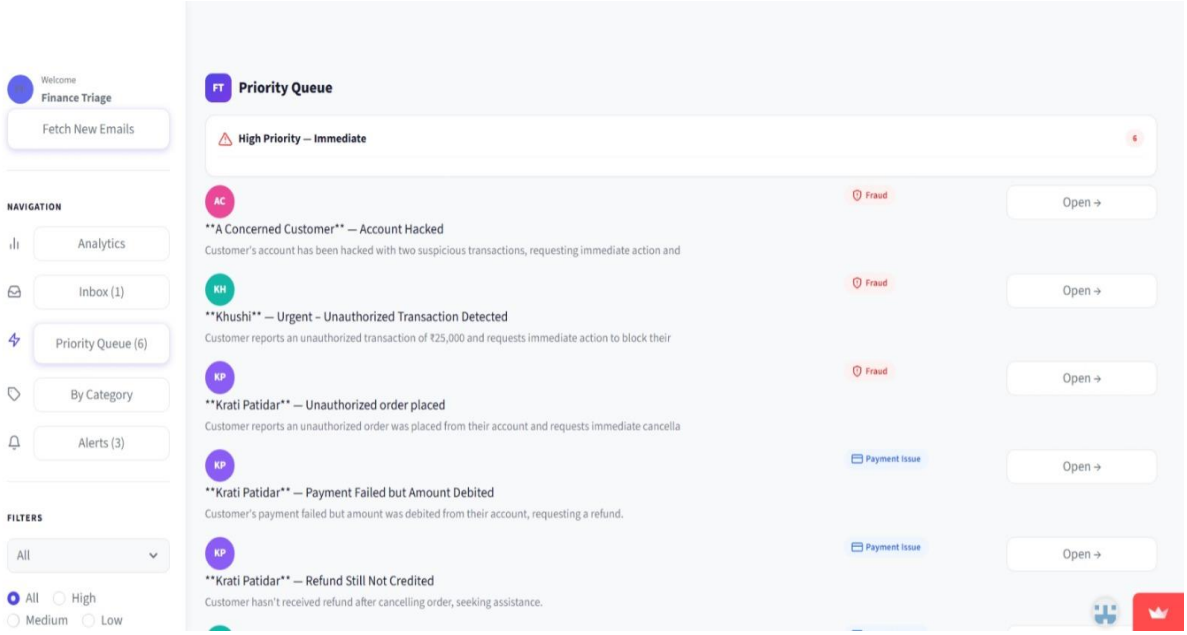


Figure 4:

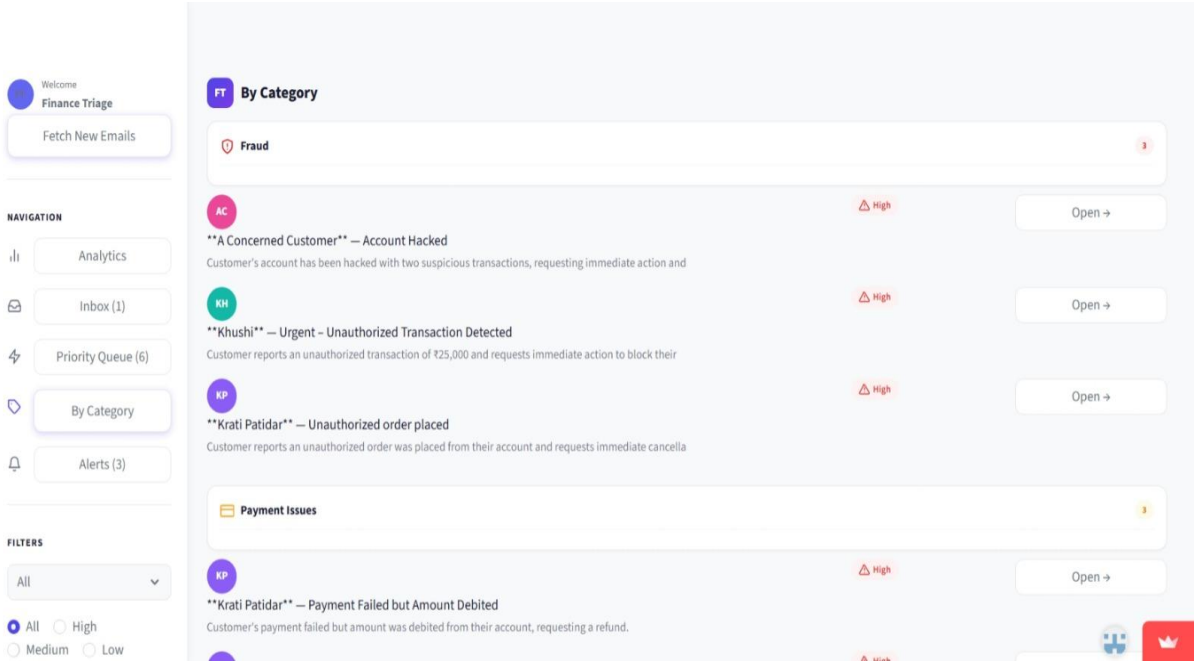


Figure 5:

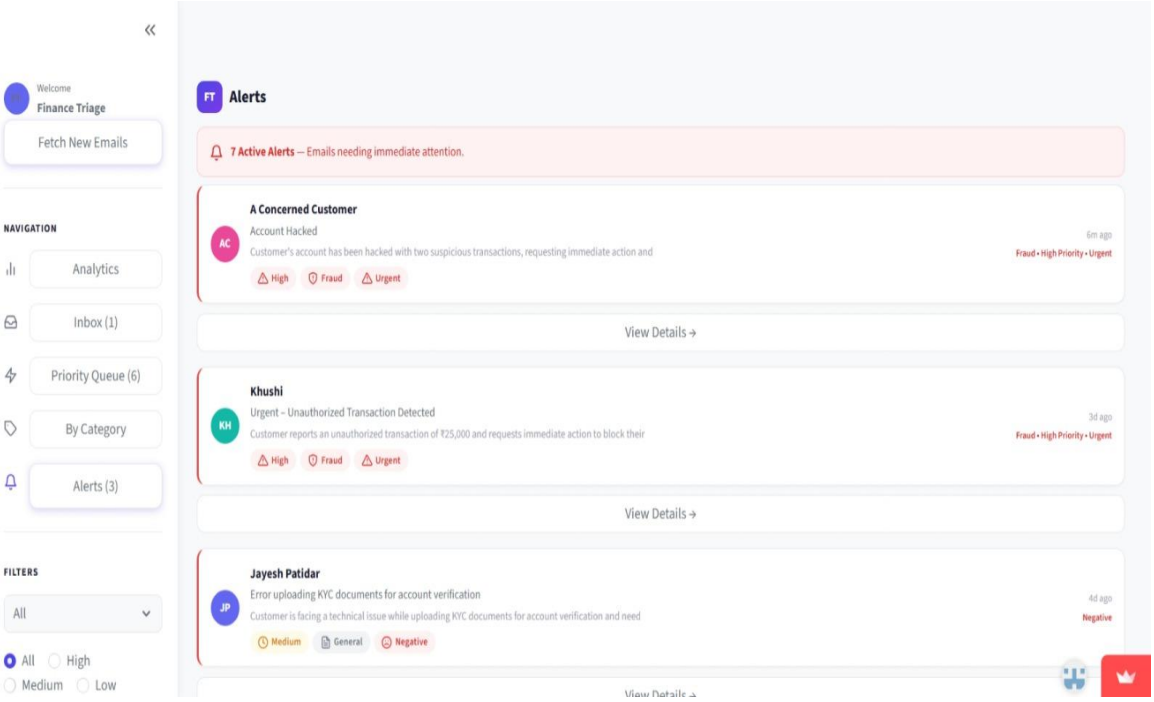
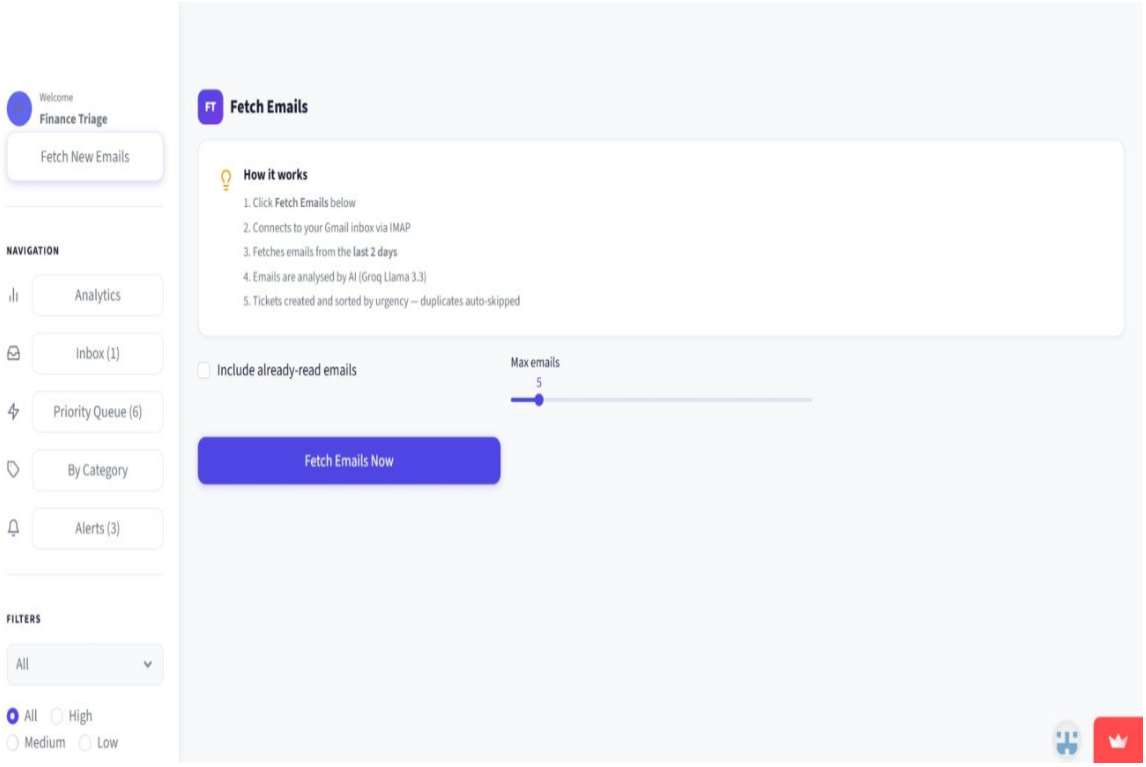


Figure 6:



Reports / Dashboards / Models:
• Streamlit dashboard for ticket management.

Key Outcomes:
• Automated triage reduces manual effort.
• Faster response generation using draft replies.
• Structured storage enables analytics and tracking.

Conclusion

The Finance Support Triage Agent demonstrates how Agentic AI can automate finance email workflows. By combining FastAPI, LangChain, and LLMs, the system transforms unstructured emails into actionable insights. The project enhanced skills in backend development, LLM integration, and data modeling.

Future Scope & Enhancements

- **Real-time email ingestion and deployment**
  Implement live production deployment using webhook-based or event-driven email ingestion for instant processing of incoming support emails without manual triggers.
- **Role-based dashboard with approval workflows**
  Introduce multi-level access control for agents, managers, and admins with features like approval pipelines, audit logs, and workflow tracking for enterprise readiness.
- **Multi-language support**
  Enable multilingual email understanding and response generation using advanced LLM capabilities to support global users and diverse customer bases.
- **Advanced analytics and reporting**
  Add intelligent analytics such as trend analysis, fraud pattern detection, ticket resolution metrics, and predictive insights for data-driven decision-making.
- **Integration with CRM or ticketing platforms**
  Provide seamless integration with platforms like Zendesk, Freshdesk, or Salesforce to embed the system into existing enterprise workflows.
- **Model fine-tuning for higher accuracy**
  Fine-tune the AI model using domain-specific financial datasets to improve intent classification, entity extraction, and response relevance.
- **Scalability and performance optimization**
  Optimize system performance through lightweight deployments, reduced inference latency, and cost-efficient infrastructure for large-scale adoption.
- **Enhanced security and compliance**
  Implement enterprise-grade security features such as data encryption, access control, and compliance with financial data regulations.